Schubert Calculus of Hyperplanes

based on joint works with (1) Kirill Zainoulline (arXiv:2404.07314), (2) Sergey Galkin, Naichung Conan Leung and Changzheng Li (arXiv:2509.01101), and (3) Changzheng Li, Jiayu Song and Mingzhi Yang (arXiv:2509.17857)

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Schubert meets Lefschetz

Today, I am going to talk about

hypersurfaces in homogeneous varieties.

Theorems on **homogeneous varieties** are counted as "Schubert calculus" to honor his contribution.

Theorems on **hypersurfaces** are mostly named after "Lefschetz" to honor his contribution.

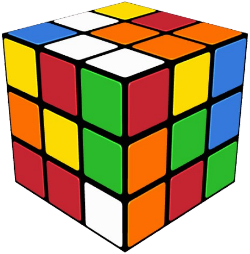
Hermann Schubert

Solomon Lefschetz

From homogeneous varieties to their hyperplanes

a homogeneous variety

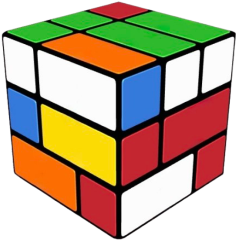
is like



a **Rubik's cube**, whose positions form a group.

a hypersurface of it

is like



a **bandaged Rubik's cube** with many moves blocked.

Hodge diamonds

Hodge systematically studied the topology of algebraic varieties. The Hodge diamond refines the Betti number of a smooth projective variety.

$$\begin{array}{ccccccc}
 & & & 1 & & & \\
 & & 0 & & 0 & & \\
 & 0 & & 1 & & 0 & \\
1 & & 101 & & 101 & & 1 \\
 & 0 & & 1 & & 0 & \\
 & & 0 & & 0 & & \\
 & & & 1 & & &
\end{array}$$

a quintic 3-fold in $\mathbb{P}^4$

$$\begin{array}{ccccccccc}
 & & & & 1 & & & & \\
 & & & 0 & & 0 & & & \\
 & & 0 & & 1 & & 0 & & \\
 & 0 & & 0 & & 0 & & 0 & \\
0 & & 0 & & 1 & & 0 & & 0 \\
 & 0 & & 0 & & 0 & & 0 & \\
 & & 0 & & 1 & & 0 & & \\
 & & & 0 & & 0 & & & \\
 & & & & 1 & & & &
\end{array}$$

projective space $\mathbb{P}^4$

W. V. D. Hodge

Remark: “quintic” means degree 5.

Hodge made the following famous conjecture

$$\text{middle column} \cap \text{cohomology} = \text{algebraic cycles}.$$

Hogde–Tate type

The homogeneous variety $X = G/P$ admits an **affine paving** (by Schubert varieties). It brings G/P with many nice properties, including

only the "middle column" of Hodge diamond of G/P is nonzero

which is known as "**Hodge–Tate**" property.

To do Schubert calculus on a hyperplane $Y \subset X$, $H^*(Y)$ should be at worst of Hogde–Tate type. The first question is,

when $Y \subset X$ is of Hodge–Tate type?

Hirzebruch provided an algorithm of computing the Hodge diamond of Y.

Friedrich Hirzebruch

Hermann Grassmann

Let us consider Grassmannians.

$$\mathrm{Gr}(k,n)=\{k\text{-subspaces } V\le\mathbb{C}^n\}.$$

The Picard number is one and thus it is okay to talk about a "hyperplane". Examples:

- $X=\mathrm{Gr}(1,n)=\mathbb{P}^{n-1}$, then $Y\simeq\mathbb{P}^{n-1}$;
- $X=\mathrm{Gr}(2,2n)$, then $Y\simeq\mathrm{SG}(2,2n)$;
- $X=\mathrm{Gr}(3,10)$, by [DV10], Y is not HT.

O. Debarre and C. Voisin, *Hyper-Kähler fourfolds and Grassmann geometry*, J. Reine Angew. Math. 649 (2010), 63–87.

$$\begin{array}{ccccccc} & & & \vdots & & & \\ & & & 20 & & & \\ \cdots & 0 & 1 & 30 & 1 & 0 & \cdots \\ & & & 20 & & & \\ & & & \vdots & & & \end{array}$$

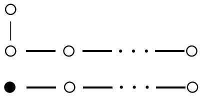
Classification

Theorem (Galkin–Leung–Li–X. 2025)

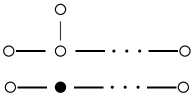
The cohomology $H^*(Y)$ is of Hodge–Tate type if and only if

- $Y \subset \mathrm{Gr}(1, n) \simeq \mathrm{Gr}(n-1, n)$ *where* $n \geq 1$;
- $Y \subset \mathrm{Gr}(2, n) \simeq \mathrm{Gr}(n-2, n)$ *where* $n \geq 4$;
- $Y \subset \mathrm{Gr}(3, n) \simeq \mathrm{Gr}(n-3, n)$ *where* $n = 6, 7, 8$.

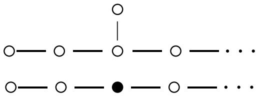
The result can be viewed as an ADE classification



$\mathbb{A}_n : \mathrm{Gr}(1, n)$



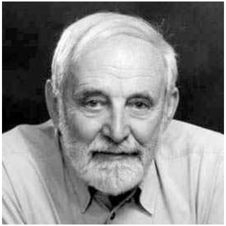
$\mathbb{D}_n : \mathrm{Gr}(2, n)$



$\mathbb{E}_n : \mathrm{Gr}(3, n)$

It reflects all exceptional isomorphisms.

Bott–Borel–Weil Theorem



Raoul Bott

Bott proved a parabolic generalization of Borel–Weil theorem. Based on that, Snow [Sn86] gave a combinatorial description of

$$H^j(Gr(k,n),\Omega^p(\ell)) \neq 0$$

in terms of partitions. We need the vanishing to use Griffiths' argument.

8	7	5	4	2	1
5	4	2	1		
2	1				

$H^6(\mathrm{Gr}(3,9),\Omega^{12}(3))$

7	5	3	1
5	3	1	
3	1		
1			

$H^6(\mathrm{Gr}(4,8),\Omega^{10}(2))$

7	6	3	2
6	5	2	1
3	2		
2	1		

$H^4(\mathrm{Gr}(4,8),\Omega^{12}(4))$

D. Snow, *Cohomology of twisted holomorphic forms on Grassmann manifolds and quadric hypersurfaces*, Math. Ann. 276.1 (1986): 159-176.

Generalized Grassmannians

When P is maximal, we call G/P_k a **generalized Grassmannians**. Many properties are included in the periodic table (QR code on the right). They all have Picard number one, so it makes sense of "hyperplane".

Conjecture (Galkin–Leung–Li–X. 2025)

The cohomology $H^*(Y)$ is of Hodge–Tate type if and only if $\dim G/P_k < 2\,\mathrm{ind}(G/P_k)$, *namely G/P_k is given by one of the cases:*

1. *a complex Grassmannian mentioned as above;*
2. *a (co)adjoint Grassmannian [BP22];*
3. (i) B_n/P_1 *or* D_n/P_1, (ii) C_3/P_3, (iii) $B_{n-1}/P_{n-1} \simeq D_n/P_n \simeq D_n/P_{n-1}$ *for* $n \in \{4, 5, 6, 7\}$, (iv) $E_6/P_1 \simeq E_6/P_6$ *or* (v) E_7/P_7.

Benedetti, V.; Perrin, N. Cohomology of hyperplane sections of (co)adjoint varieties. *arXiv:2207.02089*.

Graded Lie Algebras

We do not have a conceptual explanation of this list. But it seems that it is closely related to graded Lie algebra. For example,

$$
\begin{array}{llcc}
(1) & \mathfrak{sl}_{n+1} & \cong & \overset{-1}{V^*} \oplus \overset{0}{\mathfrak{gl}_n} \oplus \overset{1}{V} \\
& \mathfrak{so}_{2n} & \cong & \overset{-1}{\Lambda^2 V^*} \oplus \overset{0}{\mathfrak{gl}_n} \oplus \overset{1}{\Lambda^2 V} \\
& \mathfrak{e}_n & \cong & \cdots \oplus \overset{-1}{\Lambda^3 V^*} \oplus \overset{0}{\mathfrak{gl}_n} \oplus \overset{1}{\Lambda^3 V} \oplus \cdots \\
(2) & L\mathfrak{g} & \cong & \cdots \oplus \overset{-1}{\mathfrak{g}} \oplus \overset{0}{\mathfrak{g}} \oplus \overset{1}{\mathfrak{g}} \oplus \cdots \\
& L\mathfrak{g}^{tw} & \cong & \cdots \oplus \overset{-1}{V^\vee} \oplus \overset{0}{\mathfrak{g}} \oplus \overset{1}{V} \oplus \cdots
\end{array}
$$

Invariant cycle theorem

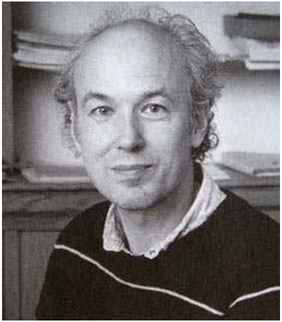
Lefschetz hyperplane theorem tells that when Y is ample, the restriction of

$$\mathrm{res} : H^*(X) \longrightarrow H^*(Y)$$

is injective for $* < \dim Y$. **Deligne invariant cycle theorem** says that

$$\mathrm{res}(H^*(X)) = H^*(Y)^{\text{monodromy}}.$$

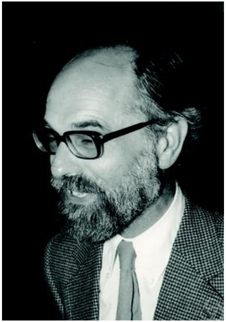
This plays a fundamental role in our proof.



Pierre Deligne

The monodromy comes from the fact that hyperplanes form a family and the smooth loci is not necessarily simply-connected.

Adjoint varieties



T. A. Springer

The monodromy in the case X is an adjoint variety can be determined relatively easily. In this case, the Plücker embedding

$$X = G/P \longrightarrow \mathbb{P}(\mathfrak{g}).$$

Thus a hyperplane is determined by an element of $\mathfrak{g}^* \simeq \mathfrak{g}$. In this case, applying Fourier–Deligne transformation on perverse sheaves, we can determine the monodromy via **Springer theory**.

Precisely, in this case, the smooth loci of $\mathfrak{g}$ is exactly

$$\mathfrak{g}^{\mathrm{rs}} = \{\text{regular semisimple elements}\}$$

whose fundamental group is the **braid group**.

Application A

The monodromy action also exists for quantum cohomology, and we use it to argue the vanishing of Gromov–Witten invariants.

Theorem (Galkin–Leung–Li–X. 2025, Conjectured by [BP22])

Let Y be a smooth hyperplane section of G/P of (co)adjoint type of type D or type E. Then $QH^*(Y)$ is not semi-simple.

The main reason that it is not applicable to type A:

Lemma

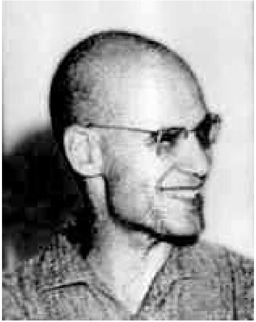
For finite irreducible Coxeter group W not of type A, $\dim(\mathrm{Sym}^3 \chi_{\mathrm{std}})^W = 0$.

Benedetti, V.; Perrin, N. Cohomology of hyperplane sections of (co)adjoint varieties. *arXiv:2207.02089*.

Application B

Motivic decomposition achieved success in the study of quadratics and algebraic groups (over non-algebraically closed fields). The idea is to decompose twisted homogeneous varieties in the category of motives, i.e. the **Karoubi envelope** (idempotent completion) of the category of correspondences.

Similarly, why not study the motivic decomposition of a hyperplane?



Alexander Grothendieck

Theorem (X.–Zainoulline, 2024)

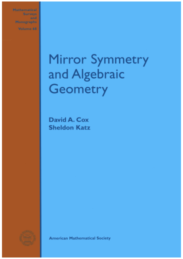
Artin motive (motive of a field extension) appears in the motivic decomposition of a hyperplane of a twisted Milnor hypersurface (adjoint variety of type A).

Quantum cohomology

The structure of cohomology $H^*(Y)$ is almost determined by the aforementioned approaches. Then let us turn to **quantum cohomology**.

Quantum cohomology is a deformation of cohomology ring, packaging many information of Gromov–Witten invariants (curve counting) together.

The quantum cohomology of homogeneous variety G/P is well understood nowadays thanks to the efforts of a generation of mathematicians (from 1997).



Mirror Symmetry and Algebraic Geometry

Functoriality

Quantum cohomology is no longer functorial (at least naïvely):

$$\begin{array}{ccc} \text{morphism} & & \text{ring homomorphism} \\ Y \longrightarrow X & \not\Longrightarrow & QH^*(X) \longrightarrow QH^*(Y). \end{array}$$

It is still not clear how much we can talk about the functoriality of quantum cohomology. Luckily, in the case $Y \subset X$ is a hyperplane, the correct functoriality is graduately understood in

- There is a Lefschetz theorem for quantum cohomology [CG07].
- We are guided by quantum Lefschetz hyperplane principle [GI].

T. Coates and A. Givental, *Quantum Riemann-Roch, Lefschetz and Serre*, Ann. of Math. (2) 165 (2007), no. 1, 15–53.

S. Galkin and H. Iritani, *Quantum ring and quantum Lefschetz*, in preparation.

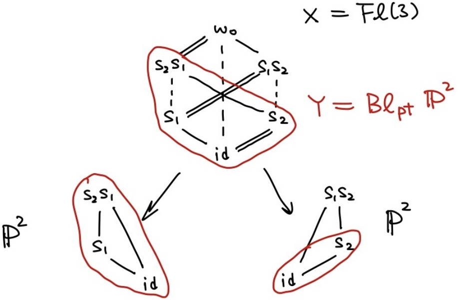
Positivity and Virtuality

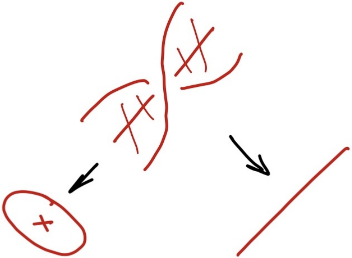
Since the tangent bundle of G/P is globally generated, the variety G/P is convex. It implies

$$[\overline{\mathcal{M}}_{0,3}(G/P,d)]^{\mathrm{vir}} = [\overline{\mathcal{M}}_{0,3}(G/P,d)].$$

With Kleiman transversality theorem, quantum coefficients can be translated to honest curve counting, and thus positive.

However, these are no longer true for a hyperplane $Y \subset X$.





Chevalley–Pieri rules

Theorem (Galkin–Leung–Li–X. 2025)

For a Hogde–Tate hyperplane $j : Y \subset \mathrm{Gr}(k,n)$, we have

$$j^*\sigma_{1^p} \star j^*\sigma_\lambda = j^*(\textit{classical terms}) + j^*(\textit{quantum terms of } \sigma_{1^p} \star (\sigma_\lambda \cup \sigma_1)).$$

Note that $\deg q_Y = \dim q_X - 1$.

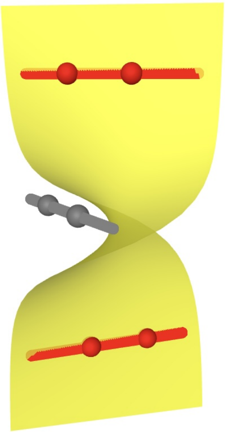
Theorem (Li–Song–Yang–X. 2025)

Let us consider a hypersurface $j : Y \subset \mathcal{F}\ell(n)$ of degree $(0,\dots,0,1)$.

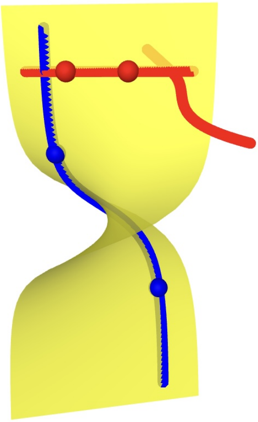
$$\begin{aligned} j^*\sigma^{s_k} \star j^*\sigma^u = {} & j^*(\textit{terms without } q_{n-1}) \\ & + j^*(\textit{terms with } q_{n-1} \textit{ in } \sigma^{s_k} \star (\sigma^u \cup \sigma^{s_{n-1}})) \\ & - \delta_{k,n-1} j^*\sigma^u \cdot q_{n-1}. \end{aligned}$$

Note that $\deg q_1 = \cdots = \deg q_{n-2} = 2$ *and* $\deg q_{n-1} = 1$.

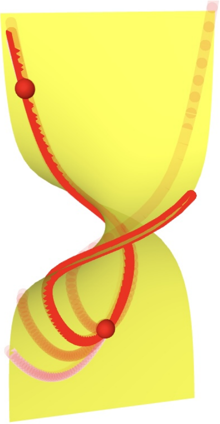
Behavior of Curves



degree = 0



degree = 1



degree $\geq$ 2

Quantum Schubert Polynomials

In the second theorem, we considered a hypersurface $j : Y \subset \mathcal{F}\ell(n)$ of degree $(0, \ldots, 0, 1)$. Note that Y is a smooth Schubert variety of $w_0 s_{n-1}$. We have the following a little bit surprising result

Theorem (Li–Song–Yang–X. 2025)

The class $j^*\sigma^w$ is represented by the quantum Schubert polynomial $\mathfrak{S}^q_w$.

Note that this does not imply the naïve map $j^* : QH^*(X) \to QH^*(Y)$ is a ring homomorphism. Our proof is based on the transition formula noticed and applied early in [LOTRZ24].

T. Le, S. Ouyang, L. Tao, J. Restivo and A. Zhang, *Quantum bumpless pipe dreams*, Forum Math. Sigma 13 (2025).

Thank You

and here is a happy plane